# Bernardo Camajori Tedeschini

AI/ML scientist | PhD | ML for Positioning & Wireless

@ bernicama@gmail.com  Residence : Italy  bernardocama.github.io

github.com/BernardoCama  Google Scholar  linkedin.com/in/bernardo-camajori-tedeschini-ba365918b


## Work experience, stages, studies abroad

**Today – August 2026** — AI/ML Scientist Staff, NATO CMRE, La Spezia, Italy

- Development of AI/ML algorithms for maritime surveillance at the NATO Centre for Maritime Research and Experimentation, a NATO Centre of Excellence (formerly SACLANT, Supreme Allied Commander Atlantic).
- Member of the Data Knowledge and Operational Effectiveness (DKOE) team.

AI/ML | Maritime Surveillance | Deep Learning | Data Fusion | Python

**August 2026 – February 2025** — Senior Systems Engineer, Qualcomm Inc., France

- Wireless R&D for 6G Non-Terrestrial Networks (NTN) : system design, link-level simulations, and algorithm development for next-generation positioning and communication systems.
- GNSS-independent positioning research and AI-enhanced RAN Intelligent Controller (RIC) design.
- Filed 40+ patent applications and contributed to pre-standardisation activities aligned with 3GPP RAN working groups.

5G NR | 6G | Python | MATLAB | C/C++ | 3GPP | System Simulation

**August 2024 – August 2023** — Visiting Researcher at MIT Laboratory for Information & Decision Systems (LIDS), Massachusetts Institute of Technology, Cambridge

- Development of classical and deep machine learning algorithms for localization, ranging, and sensing networks. Research carried out under Prof. Win's supervision.
- Designed real-time Bayesian Neural Networks for cooperative positioning and tracking, achieving sub-meter accuracy in 6G scenarios.
- Implemented multi-agent reinforcement learning for distributed cooperative positioning.
- System-level simulations using MATLAB and Python; ray-tracing channel modeling.

Bayesian Neural Networks | Ranging & Positioning | Signal Processing | Ray-tracing | 5G/6G | MATLAB | Python

**July 2021 – June 2021** — Visiting Research Scientist, CERN, Geneva

- Designed and deployed the full network architecture for privacy-preserving Federated Learning across physically separated machines at CERN, Milan, and Athens.
- On-device distributed training with heterogeneous data and network conditions.

Federated Learning | Privacy-preserving ML | Distributed Systems | NVIDIA | Docker

**July 2013 – June 2013** — Summer english school, University of Limerick, Ireland

- Contribution : Attending daily lessons of english and culture of Ireland.

## Academics

**November 2024 – November 2021** — PhD in Information Technology (Honors - Cum Laude), Telecommunications, Politecnico di Milano

- Thesis : Cooperative Machine Learning Methods in Distributed Systems.
- Multi-agent learning for cooperative positioning and sensing.
- On-device ML inference and TinyML for efficient edge computing.
- Federated and distributed learning in privacy-constrained networks.

**October 2021 – October 2019** — Master Degree in Telecommunication Engineering (Honors - 110L, GPA 29.85/30), Communication Networks and Internet, Politecnico di Milano

- Thesis in collaboration with CERN, CNR, and medical school of Athens.
  Title : "Federated learning architectures and algorithms for diagnostic imaging in healthcare networks."

**July 2019 – October 2016** — Bachelor Degree in Computer Science Engineering (Honors - 110L, GPA 29.23/30), Telecommunications, Politecnico di Milano

- Thesis : Final Examination in Communication Systems.

## Honors and Awards

2026 Best PhD Theses Award, IEEE Aerospace and Electronic Systems Society (AESS) Italy Chapter.
2025 Prof. Florian Daniel Award for the best PhD thesis in Information Technology, Politecnico di Milano.
2025 Chorafas Award by the Dimitris N. Chorafas Foundation to the best PhD students of each partner university.
2025 Springer Award for an outstanding PhD dissertation in Information Technology, Politecnico di Milano.
2024 PhD degree with Honors.
2022 Roberto Rocca Doctoral Fellowship : funding for 12 months of research at MIT.
2021 PhD grant from Ministero dell'Istruzione, dell'Università e della Ricerca (MIUR), Italy.
2021 Master's degree with Honors.
2019 Bachelor's degree with Honors.
2017 Best freshmen of the academic year 2016/2017, awarded by Politecnico di Milano.

## Projects

### 6G-NTN
2023 - 2025

6g-ntn.eu

- Partners : University of Bologna, Thales Alenia Space, Thales SIX, Ericsson, Orange, SES Techcom, DLR, CTTC, Qualcomm, and others.
- Objective : EU SNS JU (Horizon Europe) RIA to design and validate technical, regulatory, and standardisation enablers for full integration of non-terrestrial networks with terrestrial 6G.
- Budget : 5.1M €. Call : HORIZON-JU-SNS-2022-STREAM-B-01-03 (GA No. 101096479).
- Contribution : GNSS-independent positioning algorithms, AI-enhanced RAN Intelligent Controller (RIC), and pre-standardisation support aligned with 3GPP activities.

PyTorch | MATLAB | 3GPP | Positioning | vLEO/GEO/HAPS/UAV | D2D

### TRUSTroke
2022 - 2025

trustroke.eu

- Partners : Fundacio Hospital Universitari Vall D'Hebron, Fondazione Policlinico Universitario Agostino Gemelli-IRCSS, KU Leuven, CERN, Eurecat Technology Centre, Nora Health, Politecnico di Milano, CNR, Stroke Alliance for Europe and others.
- Objective : Trustworthy prediction of stroke outcome on a Federated Learning infrastructure.
- Budget : 6M €. Call : HORIZON-HLTH-2022-STAYHLTH-01-two-stage.
- Contribution : Analysis of the federated algorithms and their optimization.

Tensorflow | Fully-distributed Federated Learning | Asynchronous Federated Learning

### HYPER-5G
2022 - 2024

hyper5g-project.eu

- Partners : Geomatics Research & Development srl, algoWatt SpA and Politecnico di Milano.
- Budget : 500k € funded by ESA's Navigation Innovation and Support Program (NAVISP).
- Objective : Studying, designing and developing the algorithms and software needed to implement a precise positioning engine to jointly use multi-constellation GNSS and 5G observations.
- Contribution : Design and implement the 5G positioning system, including first-path-of-arrival estimation and ranging algorithms.
- Lab measurements with commercial 5G equipment and signal analyzers.

MATLAB | Rohde & Schwarz | 5G Positioning | GNSS | Ranging

### CAFEIN-FL
2019 - 2021

kt.cern/kt-fund/projects/cafein-federated-network-platform-development-and-deployment-ai-based-analysis-and

- Partners : CERN, CNR, Politecnico di Milano, National and Kapodistrian University of Athens.
- Budget : 135k CHF funded by CERN Knowledge Transfer fund.
- Objective : Federated network platform for the development and deployment of AI based analysis and prediction models.
- Contribution : Creation of the whole network architecture and proof-of-work implementation of Federated Learning algorithms.

Tensorflow | Personalized Federated Learning

## Certifications

| | |
|---|---|
| October 2025 | PRINCE2 Practitioner |
| September 2025 | PRINCE2 Foundation |
| February 2025 | Qualcomm Technologies 5G Associate-Level Certification |
| October 2022 | Professional engineering license |
| July 2022 | MIT Certification of English Proficiency Assessment |
| November 2021 | TOEFL |
| October 2021 | RTF Certificate (VHF) |
| August 2021 | First Aid Certificate |
| September 2018 | ETS - TOEIC (Test of English for International Communication - Listening and Reading Test ), Grade 835 (C1) |
| May 2015 | Certificate in ESOL International (First), Grade C |

## Technical Skills

| | |
|---|---|
| **Programming Languages** | Python (Advanced), MATLAB (Advanced)<br>C/C++ (Intermediate) |
| **ML Frameworks** | PyTorch, TensorFlow, Keras |
| **Wireless Standards** | 5G NR, LTE, 3GPP, IEEE 802.15.4z (UWB)<br>IEEE 802.11az/ax/be, Bluetooth/BLE |
| **Tools & Platforms** | git, Linux, Docker, NVIDIA CUDA |
| **Lab & Measurement** | Rohde & Schwarz, Spectrum Analyzers |
| **System Design** | Link/System Simulations, Ray-tracing |

## Interests

- Ranging & Positioning
- ML for Wireless Systems
- Cooperative & Multi-agent Systems
- On-device / Low-power ML
- Spatial Computing & Sensing

## Languages

| | |
|---|---|
| Italian | Native |
| English | Fluent (C1–C2) |
| French | Intermediate (B1) |

## Soft Skills

- Problem Solving
- Team Working
- Leadership
- Curiosity and innovation
- Organization and planning

## References

**Monica Barbara Nicoli**
*Associate Professor at Polimi*
Milan, Italy
@ monica.nicoli@polimi.it

**Luigi Serio**
*High-ranking official at CERN*
Geneva, Switzerland
@ luigi.serio@cern.ch

**Moe Z. Win**
*Core faculty Professor at MIT*
Cambridge, MA, USA
@ moewin@mit.edu

**Amira Alloum**
*Director of Engineering at Qualcomm*
Paris, France
@ aalloum@qti.qualcomm.com

**Paolo Braca**
*Head of Branch at NATO CMRE*
La Spezia, Italy
@ paolo.braca@cmre.nato.int

## Publications

Bernardo Camajori Tedeschini has co/authored **more than 20 publications**, including research articles in high-impact peer-reviewed journals such as *IEEE Communications Surveys & Tutorials* (I.F. 34.4), *IEEE Journal on Selected Areas in Communications* (I.F. 13.8), *IEEE Transactions on Intelligent Vehicles* (I.F. 14), *IEEE Transactions on Network Science and Engineering* (I.F. 6.7), *IEEE Transactions on Signal Processing* (I.F. 4.6), and *Scientific Reports* (Nature portfolio, I.F. 3.8). His current *h*-index is 13, and his work has garnered more than **600 citations** (Scopus).

## Journal Papers

May 2026 P. Xu, M. Brambilla, **B. Camajori Tedeschini**, and M. Nicoli, "Experimental Validation of 5G Positioning With Inter-Cell Clock Bias Correction", IEEE Access, doi : 10.1109/ACCESS.2026.3694117.

Jan 2025 **B. Camajori Tedeschini**, S. Savazzi, M. Nicoli, "Weighted Consensus Algorithms in Distributed and Federated Learning", IEEE Transactions on Network Science and Engineering (TNSE), doi : 10.1109/TNSE.2025.3528982.

Nov 2024 M. Brambilla, M. Alghisi, **B. Camajori Tedeschini**, A. Fumagalli, F. Grec, L. Italiano, C. Pileggi, L. Biagi, S. Bianchi, A. Gatti, A. Goia, M. Nicoli, and E. Realini, "Integration of 5G and GNSS Technologies for Enhanced Positioning : an Experimental Study", IEEE Open Journal of the Communications Society (OJ-COMS), doi : 10.1109/OJCOMS.2024.3487270.

Oct 2024 **B. Camajori Tedeschini**, M. Brambilla, M. Nicoli, and M. Z. Win, "Multi-agent Reinforcement Learning for Distributed Cooperative Positioning", IEEE Transactions on Intelligent Vehicles (TIV), doi : 10.1109/TIV.2024.3471909.

Sep 2024 L. Italiano, **B. Camajori Tedeschini**, M. Brambilla, H. Huang, M. Nicoli and H. Wymeersch, "A Tutorial on 5G Positioning", IEEE Communications Surveys & Tutorials (COMST), doi : 10.1109/COMST.2024.3449031.

Sep 2024 **B. Camajori Tedeschini**, G. Kwon, M. Nicoli, and M. Z. Win, "Real-time Bayesian Neural Networks for 6G Cooperative Positioning and Tracking", IEEE Journal on Selected Areas in Communications (JSAC), special issue "Positioning and Sensing Over Wireless Networks", doi : 10.1109/JSAC.2024.3413950.

Aug 2024 **B. Camajori Tedeschini**, M. Brambilla and M. Nicoli, "Split Consensus Federated Learning : an Approach for Distributed Training and Inference", IEEE Access, doi : 10.1109/ACCESS.2024.3446577.

Mar 2024 L. Barbieri, **B. Camajori Tedeschini**, M. Brambilla and M. Nicoli, "Deep Learning-based Cooperative LiDAR Sensing for Improved Vehicle Positioning", IEEE Transactions on Signal Processing (TSP), doi : 10.1109/TSP.2024.3377375.

Nov 2023 S. Roger, M. Brambilla, **B. Camajori Tedeschini**, C. Botella-Mascarell, M. Cobos and M. Nicoli, "Deep-Learning-Based Radio Map Reconstruction for V2X Communications", IEEE Transactions on Vehicular Technology (TVT), doi : 10.1109/TVT.2023.3326935.

Sep 2023 **B. Camajori Tedeschini**, M. Brambilla, L. Italiano, S. Reggiani, D. Vaccarono, M. Alghisi, L. Benvenuto, A. Goia, E. Realini, F. Grec and M. Nicoli, "5G positioning : a feasibility analysis with current network deployment", Scientific Reports (Nature), doi : 10.1038/s41598-023-42426-1.

Sep 2023 **B. Camajori Tedeschini** and M. Nicoli, "Cooperative Deep-Learning Positioning in mmWave 5G-Advanced Networks", IEEE Journal on Selected Areas in Communications (JSAC), special issue "5G/6G Precise Positioning on Cooperative Intelligent Transportation Systems (C-ITS) and Connected Automated Vehicles (CAV)", doi : 10.1109/JSAC.2023.3322795.

Aug 2023 **B. Camajori Tedeschini**, M. Brambilla and M. Nicoli, "Message Passing Neural Network Versus Message Passing Algorithm for Cooperative Positioning", IEEE Transactions on Cognitive Communications and Networking (TCCN), doi : 10.1109/TCCN.2023.3307953.

Aug 2023 **B. Camajori Tedeschini**, M. Brambilla, L. Barbieri, G. Balducci and M. Nicoli, "Cooperative lidar sensing for pedestrian detection : data association based on message passing neural networks", IEEE Transactions on Signal Processing (TSP), doi : 10.1109/TSP.2023.3304002.

May 2023 **B. Camajori Tedeschini**, M. Nicoli and M. Z. Win, "On the Latent Space of mmWave MIMO Channels for NLOS Identification in 5G-Advanced Systems", IEEE Journal on Selected Areas in Communications (JSAC), special issue "3GPP Technologies : 5G-Advanced and Beyond", doi : 10.1109/JSAC.2023.3273769.

May 2023 **B. Camajori Tedeschini**, S. Savazzi, M. Nicoli, "A Traffic Model based Approach to Parameter Server Design in Federated Learning Processes", IEEE Communications Letters (COMML), doi : 10.1109/LCOMM.2023.3272844.

Jan 2022 **B. Camajori Tedeschini**, S. Savazzi, R. Stoklasa, L. Barbieri, I. Stathopoulos, M. Nicoli, L. Serio, "Decentralized Federated Learning for Healthcare Networks : A Case Study on Tumor Segmentation", IEEE Access, doi : 10.1109/ACCESS.2022.3141913.

## Conference Papers

Sep 2026 **B. Camajori Tedeschini**, M. I. Gurelli, J. Lianghai, A. Karthik, and Q. Wu, "Incorporate Input Sensor Uncertainties in Neural Networks", IEEE 37th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), pp. 1-6, doi : 10.1109/PIMRC69400.2026.11711721.

Nov 2024 J. C. Morrison, N. Schatz, S. Kim, G. Kwon, **B. Camajori Tedeschini**, V. Weerackody, A. Conti, and M. Z. Win, "Sidelink-Enabled Cooperative Localization for xG Non-Terrestrial Networks", IEEE Military Communications Conference (MILCOM), pp. 1-6, doi : 10.1109/MILCOM61039.2024.10773683.

Nov 2024 N. Schatz, S. Kim, G. Kwon, **B. Camajori Tedeschini**, M. Ricard, T. Klein, V. Weerackody, A. Conti, and M. Z. Win, "Location Verification in Next-Generation Non-Terrestrial Networks", IEEE Military Communications Conference (MILCOM), pp. 1-6, doi : 10.1109/MILCOM61039.2024.10774026.

Jul 2024 **B. Camajori Tedeschini**, M. Brambilla, M. Nicoli, and M. Z. Win, "Cooperative Positioning with Multi-Agent Reinforcement Learning", 27th IEEE International Conference on Information Fusion (FUSION), doi : 10.23919/FUSION59988.2024.10706524.

Jun 2024 L. Italiano, **B. Camajori Tedeschini**, M. Brambilla and M. Nicoli, "Pedestrian Positioning in Urban Environments with 5G Technology", IEEE 22th Mediterranean Communication and Computer Networking Conference (MedComNet), doi : 10.1109/MedComNet62012.2024.10578126.

Jun 2024 U. Milasheuski, L. Barbieri, **B. Camajori Tedeschini**, M. Nicoli, and S. Savazzi, "On the Impact of Data Heterogeneity in Federated Learning Environments with Application to Healthcare Networks", IEEE Conference on Artificial Intelligence, pp. 1-6, doi : 10.1109/CAI59869.2024.00185.

Jun 2024 **B. Camajori Tedeschini**, G. Kwon, M. Nicoli, and M. Z. Win, "Empowering 6G Positioning and Tracking with Bayesian Neural Networks", IEEE International Conference on Communications (ICC), pp. 1-7, doi : 10.1109/ICC51166.2024.10622691.

Jun 2023 L. Barbieri, **B. Camajori Tedeschini**, M. Brambilla and M. Nicoli, "Implicit vehicle positioning with cooperative lidar sensing", 48th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1-5, doi : 10.1109/ICASSP49357.2023.10094864.

Jul 2022 **B. Camajori Tedeschini**, M. Brambilla, L. Barbieri and M. Nicoli, "Addressing data association by message passing over graph neural networks", 25th IEEE International Conference on Information Fusion (FUSION), pp. 01-07, doi : 10.23919/FUSION49751.2022.9841233.